

Summer Research Fellowship Programme (SRFP)
Jointly Sponsored by Indian Academy of Sciences, Bangalore
Indian National Science Academy, New Delhi
The National Academy of Sciences, India, Prayagraj

Summer Internship Report

Mathematical Modeling of Financial Risk using Stochastic Calculus

Submitted by:

Name: Dhwanil Shah

Institute: Indian Institute of Technology (BHU), Varanasi

Department: Mathematics and Computing

Email: dhwanils.shah.mat23@itbhu.ac.in

Under the Guidance of:

[Prof. K.S. Mallikarjuna Rao](#)

Department of Industrial Engineering and Operations Research

Indian Institute of Technology, Bombay

Duration: May – July 2025

Contents

1	Risk Measures	1
1.1	Coherent Measures of Risk	1
1.2	Value at Risk (VaR)	1
1.3	C-VaR / Expected Shortfall (ES)	1
1.4	Spectral Risk Measure	2
1.5	Marginal, Incremental & Component Measures	2
1.6	Calculating VaR/CVaR	2
2	Brownian Motion	3
2.1	Non-Differentiability	3
2.2	Quadratic Variation	3
3	Itô Calculus	3
3.1	Derivation of Itô's Lemma	3
4	Stochastic Differential Equations (SDEs)	4
4.1	Wiener Process & SDE Definition	4
4.2	Definition of Stochastic Integral	5
4.3	Examples of SDE in Finance	5
5	SDE Numerical Solutions	6
5.1	Euler–Maruyama Method	7
5.2	Convergence Analysis	7
5.2.1	SDE Simulation Standard Monte Carlo Error Analysis	8
5.2.2	Computational Cost of SDE Monte Carlo Simulation	9
6	Multilevel Monte Carlo (MLMC)	9
6.1	Paley–Wiener Representation of Brownian Motion	9
6.2	Motivation & Algorithm	10
6.3	Variance & Cost Analysis	11
7	CVaR Calculation Algorithm	13

Appendix

A	Greeks	14
A.1	Linear vs Non-Linear Products	14
A.2	Delta, Gamma, Vega, Theta, Rho	15
A.3	Taylor Series Expansion for Portfolio Value	17
B	Valuation vs Scenario Analysis	17
C	Volatility	17
C.1	Risk-Neutral Valuation	18
C.2	Causes of Volatility	18
C.3	Distribution of Daily Returns	19
C.4	Power Laws	19
C.5	Volatility Estimation	20
C.5.1	Historical	20
C.5.2	EWMA	21
C.5.3	GARCH	22
C.5.4	Mean Reversion	22
C.5.5	Maximum Likelihood & Targeting	23

Risk Measures

Coherent Measures of Risk

A risk measure is considered **coherent** if it satisfies the following properties:

- **Monotonicity:** If a portfolio produces a worse result than another portfolio for every state of the world, its risk measure should be greater.
- **Translation Invariance:** If an amount of cash K is added to a portfolio, its risk measure should decrease by K .
- **Homogeneity:** Scaling the size of a portfolio by a factor λ (while keeping the relative amounts of different items the same) should scale the risk measure by λ .
- **Sub-Additivity:** The weighted risk measure for two portfolios after merging should be no greater than the sum of their individual risk measures before merging.

Value at Risk (VaR)

- VaR answers: “We are X percent certain that we will not lose more than V dollars in time T .”
- VaR is widely used for risk management and regulatory purposes.
- However, VaR is not always sub-additive in nature, thus not abiding by the fourth property of a coherent risk measure. (Also, convexity is strong property assumed to solve optimisation problems, but VaR is not always convex in nature)

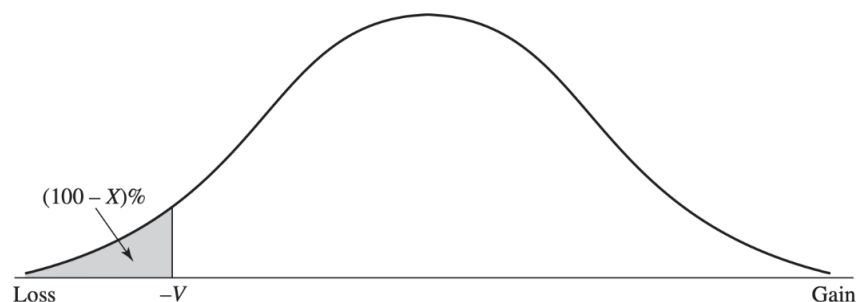


Fig: Calculation of VaR from the Probability Distribution of the Gain in the Portfolio Value. Losses are negative gains; confidence level is $X\%$; VaR level is V .

- VaR does **not** answer the question: “How bad can things get?” It only provides a threshold loss at a given confidence level, not the magnitude of extreme losses beyond that threshold.

C-VaR / Expected Shortfall (ES)

- **Expected Shortfall (ES)** is the average amount of money lost in time T if the loss exceeds a threshold V dollars.
- ES overcomes the convexity problem of Value at Risk (VaR).
- For normally distributed losses with mean μ and standard deviation σ , ES at confidence level X is:

$$ES = \mu + \sigma \frac{e^{-Y^2/2}}{\sqrt{2\pi}(1-X)}$$

where Y is the quantile corresponding to confidence level X .

Spectral Risk Measure

- A risk measure can be described by the weights it assigns to percentiles of the loss distribution.
- **VaR** gives all weight to a single percentile (the X th), while **ES** averages over all percentiles above the X th.
- **Spectral risk measures** generalize this by assigning non-decreasing weights to higher percentiles, allowing for more flexible and coherent risk assessment.
- An example is the *exponential spectral risk measure*, where weights increase exponentially with the percentile.
- Spectral risk measures are always coherent, meaning they satisfy properties like subadditivity and monotonicity.

Marginal, Incremental & Component Measures for a Portfolio

- **Marginal VaR:** Measures the sensitivity of the portfolio's Value at Risk (VaR) to the amount invested in the i th sub-portfolio. It shows how much the overall VaR would change if the investment in that sub-portfolio is marginally increased.
- **Incremental VaR:** The difference in VaR when the i th sub-portfolio is included versus when it is excluded. It quantifies the additional risk contributed by adding the sub-portfolio to the overall portfolio.
- **Component VaR:** Calculated as the product of the marginal VaR and the value of the i th sub-portfolio. It represents the portion of total VaR attributable to each sub-portfolio.

The mathematical expressions for these measures are:

$$\text{Marginal VaR}_i = \frac{\partial \text{VaR}}{\partial x_i}$$

$$\text{Component VaR}_i = \frac{\partial \text{VaR}}{\partial x_i} \cdot x_i$$

where x_i is the value invested in the i th sub-portfolio.

Calculating VaR/CVaR

- **Historical Method:** Uses actual historical returns to estimate the distribution of portfolio returns and determine VaR/CVaR.
- **Variance-Covariance Method:** Assumes returns are normally distributed and uses the mean and variance-covariance matrix of returns to calculate VaR/CVaR analytically.
- **Monte Carlo Simulation:** Simulates a large number of possible future portfolio returns based on assumed statistical properties, then estimates VaR/CVaR from the simulated distribution.

Non-differentiability of Brownian Motion

- Brownian motion is continuous everywhere but almost surely nowhere differentiable.
- At any fixed time t , the upper and lower derivatives of a Brownian path are infinite:

$$\limsup_{h \rightarrow 0} \frac{B(t+h) - B(t)}{h} = +\infty, \quad \liminf_{h \rightarrow 0} \frac{B(t+h) - B(t)}{h} = -\infty$$

- This means the path is too irregular to have a finite derivative at any point.
- This property motivated the development of stochastic calculus, as standard calculus cannot be applied to Brownian paths.

Quadratic Variation of Brownian Motion

- The **quadratic variation** of a process measures the cumulative squared increments over a time interval, capturing the roughness of the path.
- For Brownian motion B_t , the quadratic variation over $[0, t]$ is defined as:

$$[B]_t = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n (B_{t_k} - B_{t_{k-1}})^2$$

where P is a partition of $[0, t]$ and the mesh $\|P\|$ tends to zero.

- For any such sequence of partitions, the quadratic variation of standard Brownian motion is:

$$[B]_t = t$$

This means the quadratic variation increases linearly with time.

- Brownian motion has **unbounded total variation** but finite quadratic variation, which is a key property used in stochastic calculus.
- The quadratic variation distinguishes Brownian motion from smooth functions and is fundamental for defining stochastic integrals and Itô calculus.

Derivation of Itô's Lemma from Taylor Series and Quadratic Variation

Taylor Series Expansion for Stochastic Processes

Let $f(S_t, t)$ be a twice-differentiable function of a stochastic process S_t and time t . The Taylor series expansion gives:

$$df = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial S} dS_t + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} (dS_t)^2 + \dots$$

Quadratic Variation of Brownian Motion

For a process S_t following:

$$dS_t = \mu dt + \sigma dB_t$$

where dB_t is the increment of standard Brownian motion, the key properties are:

- $(dt)^2 = 0$
- $dt \cdot dB_t = 0$
- $(dB_t)^2 = dt$ (from quadratic variation)

Applying to Taylor Series

Plugging dS_t into the Taylor expansion and keeping terms up to order dt :

$$df = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial S} dS_t + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} (dS_t)^2$$

Substitute $dS_t = \mu dt + \sigma dB_t$:

$$(dS_t)^2 = (\mu dt + \sigma dB_t)^2 = \mu^2 (dt)^2 + 2\mu\sigma dt dB_t + \sigma^2 (dB_t)^2$$

Using the properties above, only the last term survives:

$$(dS_t)^2 = \sigma^2 (dB_t)^2 = \sigma^2 dt$$

Itô's Lemma

Thus, the differential becomes:

$$df = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial S} dS_t + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 dt$$

Substituting dS_t again:

$$\begin{aligned} df &= \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial S} (\mu dt + \sigma dB_t) + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 dt \\ df &= \left(\frac{\partial f}{\partial t} + \mu \frac{\partial f}{\partial S} + \frac{1}{2} \sigma^2 \frac{\partial^2 f}{\partial S^2} \right) dt + \sigma \frac{\partial f}{\partial S} dB_t \end{aligned}$$

This is the result of **Itô's Lemma**, showing how the Taylor expansion and the quadratic variation of Brownian motion lead to the extra term in stochastic calculus.

Stochastic Differential Equations (SDEs) and the Wiener Process

- The Wiener process B_t is a continuous Gaussian process with independent increments, satisfying:

$$B(0) = 0, \quad E[B(t)] = 0, \quad \text{Var}[B(t) - B(s)] = t - s, \quad 0 \leq s \leq t$$

- The general form of a stochastic differential equation (SDE) driven by Brownian motion is:

$$dX_t = a(t, X_t)dt + b(t, X_t)dB_t$$

where dW_t is the increment of the Wiener process, and X_t is the process of interest.

- The integral form of the solution is:

$$X_t = X_s + \int_s^t a(r, X_r)dr + \int_s^t b(r, X_r)dB_r$$

The first integral is a standard (Riemann) integral, while the second is a stochastic (Itô) integral.

Definition of the Stochastic Integral

- To define the stochastic integral, partition the interval $[s, t]$ into subintervals $[t_r, t_{r+1}]$.
- The Itô integral is defined as the limit:

$$\int_s^t b(r, X_r) dB_r = \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} b(\tau_r, X_{\tau_r}) [B(t_{r+1}) - B(t_r)]$$

where $\tau_r \in [t_r, t_{r+1}]$, typically chosen as the left endpoint $\tau_r = t_r$.

- The value of the stochastic integral depends on the choice of τ_r because Brownian motion has unbounded variation. The Itô convention uses the beginning of each interval.

Examples of SDE in Finance

- SDEs capture both the deterministic trend (drift) and the random fluctuations (diffusion) present in financial markets.

1. Geometric Brownian Motion (GBM) – Black-Scholes Model

- Used for modeling stock prices in the Black-Scholes option pricing framework.
- The stochastic differential equation (SDE) is:

$$dS_t = rS_t dt + \sigma S_t dB_t$$

where:

- S_t : stock price at time t
- r : risk-free interest rate
- σ : volatility of the stock
- dB_t : increment of a Wiener process (Brownian motion)

2. Heston's Stochastic Volatility Model

- Models both the stock price and its stochastic volatility.
- The SDEs are:

$$\begin{aligned} dS_t &= rS_t dt + \sqrt{V_t} S_t dB_t^{(1)} \\ dV_t &= \kappa(\theta - V_t) dt + \sigma \sqrt{V_t} dB_t^{(2)} \end{aligned}$$

where:

- S_t : stock price
- V_t : variance (stochastic volatility)
- r : risk-free rate
- κ : speed of mean reversion

- θ : long-run variance
- σ : volatility of volatility
- $dB_t^{(1)}, dB_t^{(2)}$: correlated Brownian motions

3. Cox-Ingersoll-Ross (CIR) Model with Time-Dependent Parameters

- Used for modeling the short-term interest rate.
- The SDE is:

$$dr_t = \alpha[\mu(t) - r_t]dt + \sigma\sqrt{r_t}dB_t$$

where:

- r_t : short-term interest rate at time t
- α : speed of mean reversion
- $\mu(t)$: time-dependent mean level
- σ : volatility
- dB_t : increment of a Wiener process

Solving Stochastic Differential Equations

Limitations of Traditional Methods

- **Finite Difference Methods** are widely used for solving ordinary differential equations (ODEs) and partial differential equations (PDEs) by approximating derivatives with finite differences.
- However, finite difference methods **fail for stochastic differential equations (SDEs)** because:
 - SDEs contain stochastic terms (random noise) that cannot be handled by deterministic finite difference approximations.
 - The random component dW_t in SDEs requires specialized stochastic numerical methods.
 - Traditional finite difference schemes assume smooth, differentiable functions, but Brownian motion is non-differentiable.
- Finite difference methods are limited to deterministic differential equations where derivatives can be approximated using Taylor series expansions.

Monte Carlo Simulation Methods for SDEs

- **Monte Carlo simulation** is the primary approach for solving SDEs numerically.
- The method works by:
 - Discretizing the SDE using numerical schemes like Euler-Maruyama or Milstein methods.
 - Generating many random sample paths (realizations) of the stochastic process.
 - Computing statistics (mean, variance, quantiles) from the collection of sample paths.

Euler-Maruyama Method

- The **Euler-Maruyama method** is the most basic and widely used scheme for approximating SDEs.
- For an SDE: $dX_t = a(X_t, t)dt + b(X_t, t)dW_t$
- The Euler-Maruyama approximation is:

$$Y_{n+1} = Y_n + a(Y_n, t_n)\Delta t + b(Y_n, t_n)\Delta B_n$$

where ΔW_n are independent normal random variables with mean 0 and variance Δt .

Convergence of Euler–Maruyama SDE Simulations:

Definition 1 We say the discretization $\hat{X}$ has a **strong order of convergence** of $\beta > 0$ if

$$E \left[\left\| \hat{X}_{mh} - X_T \right\| \right] \leq ch^\beta$$

for some constant c and all sufficiently small h .

Definition 2 We say the discretization $\hat{X}$ has a **weak order of convergence** of $\beta > 0$ if

$$\left| E \left[f(\hat{X}_{mh}) \right] - E \left[f(X_T) \right] \right| \leq ch^\beta$$

for some constant c (possibly depending on f), all sufficiently small h , and all $f \in C_P^{2\beta+2}$ where $C_P^{2\beta+2}$ consists of functions whose derivatives of all orders up to $2\beta + 2$ are polynomially² bounded.

- **Two main types of convergence for SDE simulation:**
 - **Weak convergence** controls the error of the means.
 - **Strong convergence** controls the mean of the error.
- To prove convergence results, assume the drift f and diffusion g satisfy global Lipschitz conditions:

$$|f(x) - f(y)| \leq L|x - y|, \quad |g(x) - g(y)| \leq L|x - y|$$

for all $x, y \in R^m$ and some constant L .

- Under these conditions, for the Euler–Maruyama method:

- **Weak order 1:**

$$\sup_{0 \leq t_n \leq T} |E[X(t_n)] - E[X_n]| = O(\Delta t)$$

- **Strong order 1/2:**

$$E \left[\sup_{0 \leq t_n \leq T} |X(t_n) - X_n| \right] = O(\Delta t^{1/2})$$

- More generally, for any $m > 1$, there exists $C(m)$ such that:

$$E \left[\sup_{0 \leq t_n \leq T} |X(t_n) - X_n|^m \right] \leq C(m) \Delta t^{m/2}$$

- **Strong convergence** is sometimes called a *pathwise property*:

- For any $\epsilon > 0$, there exists a (path-dependent) random constant $K(\epsilon)$ so that, for all sufficiently small Δt :

$$\sup_{0 \leq t_n \leq T} |X(t_n) - X_n| \leq K(\epsilon) h^{1/2-\epsilon}$$

- The Euler–Maruyama method converges well under global Lipschitz assumptions but may fail if the coefficients grow super-linearly or lack regularity.

SDE Simulation & Standard Monte Carlo Error Analysis

- To approximate $E[X(T)]$ for an SDE, Monte Carlo simulation with the Euler–Maruyama method is used.
- Fix ϵ as the accuracy requirement (confidence interval half-width) and typically use a 95% confidence level.
- Let $X_N^{[s]}$ denote the Euler–Maruyama final-time approximation along the s th sampled path. The Monte Carlo estimator is:

$$a_M = \frac{1}{M} \sum_{s=1}^M X_N^{[s]}$$

- The total error in the Monte Carlo approximation a_M can be decomposed as:

$$a_M - E[X(T)] = \underbrace{a_M - E[X_N]}_{\text{statistical error}} + \underbrace{E[X_N - X(T)]}_{\text{discretization error (bias)}}$$

- Error in a_M has two components:
 - **Statistical error** ($a_M - E[X_N]$): arises from approximating the expected value using M samples. This does not depend much on the step size; it decreases as M increases. The statistical error can be described by a confidence interval of width $O(1/\sqrt{M})$.
 - **Discretization error** ($E[X_N - X(T)]$): arises because the SDE is approximated by a difference equation, i.e., the error due to discretization (finite step size Δt). The discretization error (bias) behaves like $O(\Delta t)$ by the weak convergence, so it should also be added to the confidence interval.
- **Why is the statistical error $O(1/\sqrt{M})$?**
 - For each simulated path, the terminal value $X_N^{[s]}$ is a random variable with variance σ^2 .
 - Averaging M independent realizations gives the Monte Carlo estimator:

$$a_M = \frac{1}{M} \sum_{s=1}^M X_N^{[s]}$$

- The variance of a_M is reduced by a factor of M :

$$\text{Var}[a_M] = \frac{\sigma^2}{M}$$

- The standard deviation (standard error) is therefore $\sigma/\sqrt{M}$.
- Because the confidence interval is based on this standard error (or RMSE), the statistical error scales as $O(1/\sqrt{M})$.
- This $1/\sqrt{M}$ rate comes from the properties of averaging M i.i.d. (independent, identically distributed) random variables (the law of large numbers and central limit theorem).

- For overall confidence interval of width $O(1/\sqrt{M}) + O(\Delta t)$, to achieve accuracy ϵ both $1/\sqrt{M}$ and Δt should be of order ϵ .
- Thus, M **should scale like** ϵ^{-2} and Δt **should scale like** ϵ .

Computational Cost of SDE Monte Carlo Simulation

- The computational cost of simulating SDEs with Monte Carlo can be measured by the number of drift/diffusion function evaluations or the number of random variates generated, which is roughly proportional to the total number of time steps taken across all sample paths.
- For each path, if the time step is Δt , the cost per path is proportional to $1/\Delta t$.
- For M simulated paths, the **total cost** scales like $M/\Delta t$.
- To achieve a target accuracy ϵ :
 - The number of samples M required scales like ϵ^{-2} (from the $1/\sqrt{M}$ convergence of standard error).
 - The time step Δt required scales like ϵ (since discretization bias is $O(\Delta t)$).
 - Total cost therefore scales as $M/\Delta t \sim \epsilon^{-2}/\epsilon = \epsilon^{-3}$.

Paley–Wiener Representation of Brownian Motion

- The Paley–Wiener representation expresses Brownian motion as a series expansion in terms of orthogonal sine functions and normally distributed random variables.
- The representation over the interval $[0, 2\pi]$ is:

$$W(t) = Z_0 \frac{t}{\sqrt{2\pi}} + \frac{2}{\sqrt{\pi}} \sum_{n=1}^{\infty} Z_n \frac{\sin\left(\frac{nt}{2}\right)}{n}$$

where $Z_0, Z_1, Z_2, \dots$ are independent standard normal random variables, i.e., $Z_i \sim N(0, 1)$ for $i = 0, 1, 2, \dots$

- In practice, the infinite sum is truncated, using a finite number M of terms to approximate Brownian motion.
- **Interpretation:**
 - The **early terms** (low n) in the series determine the large-scale, low-frequency structure of the path. They affect the *overall shape*.
 - The **later terms** (high n) are associated with high-frequency sine functions and contribute only fine detail (fine-scale fluctuations).

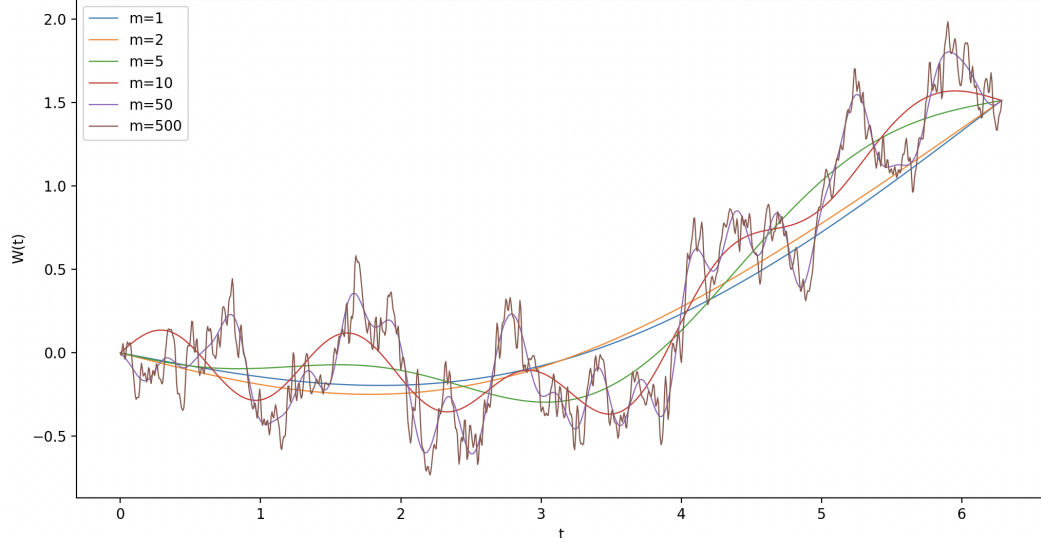


Fig: Paths based on the Paley-Wiener representation. As indicated, the six plots show the sum truncated after $M = 1, 2, 5, 10, 50$ and 500 sine terms.

- **Interpretation:** According to the image, coarse resolution paths (large Δt) account for most of the low-frequency structure, while corrections for high-frequency features (small Δt) are added sparingly and judiciously.
- This structure is closely linked to efficient simulation and multilevel Monte Carlo methods, as paths can be built incrementally by adding more terms for higher resolution.

Multilevel Monte Carlo (MLMC) Motivation and Algorithm

- **Core Idea:** Not all Monte Carlo sample paths need to be computed with the same (finest) time step—coarse, low-cost simulations suffice for low-frequency components, while fewer expensive fine-grid simulations fill in high-frequency details.
- **Variance Reduction:** Most variance in the expectation $E[f(X_T)]$ comes from the coarsest grid levels; fine grid corrections predominantly refine this estimate.
- **Algorithm Structure:**
 - **Level 0:** Calculate many simulations on a coarse grid (large Δt).
 - **Level $l > 0$:** For each finer level, simulate the differences between payoff estimates at step sizes Δt_l and Δt_{l-1} , requiring fewer paths as l increases (since variance drops rapidly with refinement).
 - **Estimator:** Total expectation is estimated by summing the average at the coarsest level and the averaged increments from each finer level:

$$E[f(X_T^{(L)})] \approx \frac{1}{N_0} \sum_{i=1}^{N_0} Y_0^{(i)} + \sum_{l=1}^L \frac{1}{N_l} \sum_{i=1}^{N_l} [Y_l^{(i)} - Y_{l-1}^{(i)}]$$

where $Y_l^{(i)} = f(X_T^{(l,i)})$ is the i -th path at level l .

Multilevel Monte Carlo with Euler–Maruyama: Core Summary

- **Goal:** Approximate $E[h(X_T)]$ for an SDE, where $h(\cdot)$ is a Lipschitz payoff, using a more efficient method than standard Monte Carlo.
- **Main Idea:** Simulate sample paths at multiple discretization levels ($l = 0, 1, \dots, L$), with stepsizes $\Delta t_l = M^{-l}T$. Combine results using a telescoping sum to reduce computational cost.
- **Estimator Construction:**

$$E[\hat{P}_L] = E[\hat{P}_0] + \sum_{l=1}^L E[\hat{P}_l - \hat{P}_{l-1}]$$

- $\hat{P}_l$ is the payoff estimated at level l using Euler–Maruyama with step Δt_l .

- **Sampling:**
 - For $E[\hat{P}_0]$, use N_0 independent coarse paths.
 - For each $E[\hat{P}_l - \hat{P}_{l-1}]$, $l > 0$, use N_l paired paths sharing the same Brownian increments for levels l and $l-1$.
- **Calculation:**

$$\hat{Y}_0 = \frac{1}{N_0} \sum_{s=1}^{N_0} \hat{P}_0^{[s]}, \quad \hat{Y}_l = \frac{1}{N_l} \sum_{s=1}^{N_l} (\hat{P}_l^{[s]} - \hat{P}_{l-1}^{[s]})$$

$$\text{Total estimator: } \hat{Y}_0 + \sum_{l=1}^L \hat{Y}_l$$

- **Computational Cost:** The MLMC method achieves a target accuracy ϵ with computational cost $O(\epsilon^{-2}(\log \epsilon)^2)$, a dramatic improvement over the $O(\epsilon^{-3})$ cost of standard Monte Carlo with uniform fine discretization.
- **Key Benefit:** MLMC attains high accuracy for SDE simulation at a fraction of the computational effort by exploiting the differing importance of computation at different scales.
- **Principle:** Most samples are run on coarse grids, where cost per path is small, while only a few samples are needed on fine grids, achieving the required accuracy for much lower overall computational effort.

Multilevel Monte Carlo: Variance and Cost Analysis—How These Results Arise

- **Variance of Estimators at Each Level:**
 - Consider the multilevel estimator for L levels:

$$\hat{Y} = \hat{Y}_0 + \sum_{l=1}^L \hat{Y}_l$$

where

$$\hat{Y}_0 = \frac{1}{N_0} \sum_{s=1}^{N_0} \hat{P}_0^{[s]}, \quad \hat{Y}_l = \frac{1}{N_l} \sum_{s=1}^{N_l} (\hat{P}_l^{[s]} - \hat{P}_{l-1}^{[s]})$$

- The variance of each term, for $l > 0$, is

$$\text{Var}[\widehat{Y}_l] = \frac{1}{N_l} \text{Var}[\widehat{P}_l - \widehat{P}_{l-1}]$$

- For a Lipschitz payoff and Euler–Maruyama method, it's known that

$$\text{Var}[\widehat{P}_l - \widehat{P}_{l-1}] = O(\Delta t_l)$$

because as the time steps halve between levels, coupled paths converge and their difference's variance decreases linearly.

- **Controlling Total Variance:**

- The overall target is mean squared error $O(\epsilon^2)$, so allocate variance equally across levels:

$$\text{Var}[\widehat{Y}_l] \leq O(\epsilon^2/L)$$

- To achieve this, choose

$$N_l = O\left(\frac{L \cdot \text{Var}[\widehat{P}_l - \widehat{P}_{l-1}]}{\epsilon^2}\right)$$

- Since $\text{Var}[\widehat{P}_l - \widehat{P}_{l-1}] = O(\Delta t_l)$, set

$$N_l = O\left(\frac{L\Delta t_l}{\epsilon^2}\right)$$

- **Cost Calculation:**

- On level l , each path costs $O(1/\Delta t_l)$ operations (number of time steps).
- The cost for level l is $N_l \cdot 1/\Delta t_l = O(L/\epsilon^2)$.
- For all L levels, the total cost is:

$$\sum_{l=0}^L N_l/\Delta t_l = (L+1)O(L/\epsilon^2) = O(L^2/\epsilon^2)$$

- Since L is chosen so that $\Delta t_L \sim \epsilon$ (to keep the bias within $O(\epsilon)$, i.e., $L = O(|\log \epsilon|)$),

$$\text{Total Cost} = O(\epsilon^{-2}(\log \epsilon)^2)$$

- **Why the Variance Decreases Linearly with Δt_l :**

- The variance of the difference between payoffs with step sizes Δt_l and Δt_{l-1} is low because the coupled paths share most Brownian increments and differ only at the finer grid.
- For Lipschitz h , $|\widehat{P}_l - \widehat{P}_{l-1}|$ is $O(\sqrt{\Delta t_l})$ in expectation, so variance is $O(\Delta t_l)$.

- **Optimizing the Work Allocation:**

- By assigning coarse grids for most samples and only a few expensive samples at the fine grids, most computation is done efficiently while still achieving the desired accuracy and variance control.

CVaR Calculation Algorithm

- The complete Conditional Value at Risk (CVaR) calculation using stochastic differential equations (SDEs) and Monte Carlo simulation involves the following steps:

1. **Initialize Parameters:** Set SDE parameters, such as drift (μ), volatility (σ), initial value (S_0), and time horizon (T).
2. **Generate Paths:** Simulate N Monte Carlo paths for the underlying process.
3. **Calculate Portfolio Values:** For each path j , compute the terminal portfolio value:

$$V_T^{(j)} = f\left(S_T^{(j)}\right)$$

where f is the portfolio value function.

4. **Compute Losses:** Calculate losses for each path:

$$L^{(j)} = \text{loss_function}\left(V_T^{(j)}\right)$$

5. **Sort Losses:** Order the losses in increasing order:

$$L^{(1)} \leq L^{(2)} \leq \dots \leq L^{(N)}$$

6. **Find VaR:** The Value at Risk (VaR) at confidence level α is:

$$\text{VaR}_\alpha = L_{(\lfloor (1-\alpha)N \rfloor)}$$

7. **Calculate CVaR:** The Conditional Value at Risk (CVaR) at level α is:

$$\text{CVaR}_\alpha = \frac{1}{\alpha N} \sum_{j=\lfloor (1-\alpha)N \rfloor}^N L^{(j)}$$

- This algorithm provides a robust way to estimate risk measures for portfolios under stochastic dynamics, accounting for both typical and extreme loss scenarios.

APPENDIX

A. Greeks

Introduction to Greeks

- Greeks are used in trading to measure different aspects of risk in a position. Traders calculate Greeks daily and must act if risk limits set by their institution are exceeded.
- Each Greek quantifies a specific risk in a trading position.
- Used for daily risk monitoring and management.

Linear Products

- A **linear product** is one whose value at any given time is linearly dependent on the value of an underlying market variable.
- **Forward contracts** are examples of linear products; **options** are not.
- Linear products are useful for hedging both large and small changes in the underlying variable. Once the hedge is set up, it does not need to be changed.

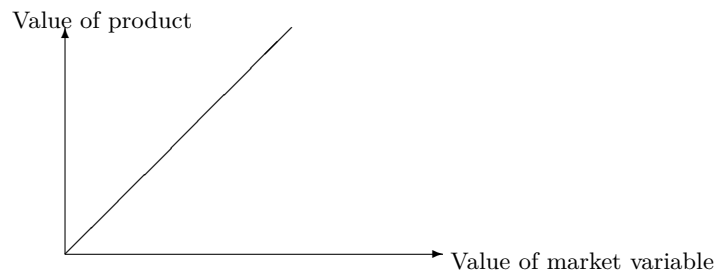


Fig: Linear Product

Non-Linear Products

- **Non-linear products** require dynamic hedging; it is not possible to hedge and forget.
- To maintain delta neutrality, the hedge must be periodically adjusted—a process called *rebalancing*.
- A key challenge with non-linear products is the transaction costs incurred due to frequent rebalancing.

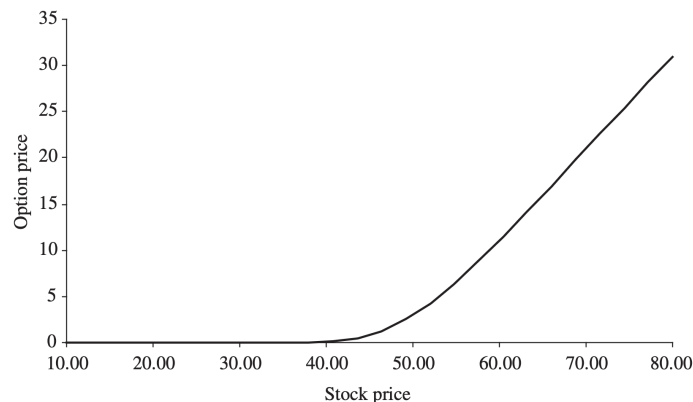


Fig: Value of a Call Option by Black-Scholes Merton's formula, a non-linear product

Delta

- Measures sensitivity of portfolio value to changes in the underlying market variable.
- Defined as:

$$\Delta = \frac{\partial P}{\partial S}$$

where P is portfolio value and S is the market variable (e.g., stock price).

- Delta hedging: Adjusting the portfolio to make it delta neutral, so small changes in the market variable do not affect portfolio value.

Gamma

- **Gamma** is the rate of change of delta with respect to the market variable (typically the underlying asset price).
- It measures the extent to which large changes in the underlying can cause problems for hedging.
- Mathematically, gamma is defined as:

$$\Gamma = \frac{\partial^2 P}{\partial S^2}$$

where P is the portfolio value and S is the market variable.

- If gamma is small, delta changes slowly, so adjustments to keep a portfolio delta neutral are needed only infrequently.
- If gamma is large, delta is highly sensitive to the underlying price, making it risky to leave a delta-neutral portfolio unchanged for long.

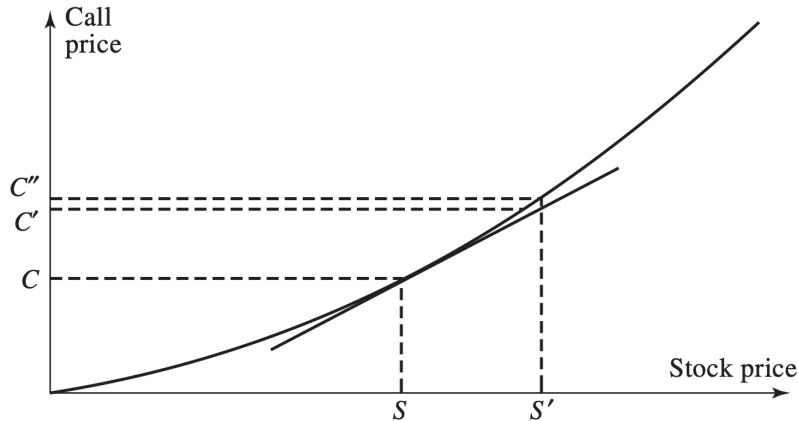


Fig: Hedging Error Introduced by Non-Linearity

When the stock price moves from S to S' , delta hedging assumes the option price moves from C to C' , but in reality it moves from C to C'' . The difference between C' and C'' leads to a hedging error. This error depends on the curvature of the relationship between option price and stock price, which is measured by gamma.

Gamma Neutral Portfolio

- A **linear product** (like a forward contract) has zero gamma and cannot be used to adjust the gamma of a portfolio.
- To manage gamma, a position in a non-linear instrument (such as an option) is required.
- Suppose a delta-neutral portfolio has a gamma of Γ , and a traded option has a gamma of Γ_T . If w_T is the number of traded options added, the total gamma of the portfolio becomes:

$$w_T \Gamma_T + \Gamma$$

- To make the portfolio gamma neutral, set the total gamma to zero:

$$w_T = -\frac{\Gamma}{\Gamma_T}$$

- Adding the traded option may change the delta of the portfolio, so the position in the underlying asset must be adjusted to maintain delta neutrality.
- Gamma neutrality is only temporary; as time passes, the position in the traded option must be continuously adjusted to keep the portfolio gamma neutral.

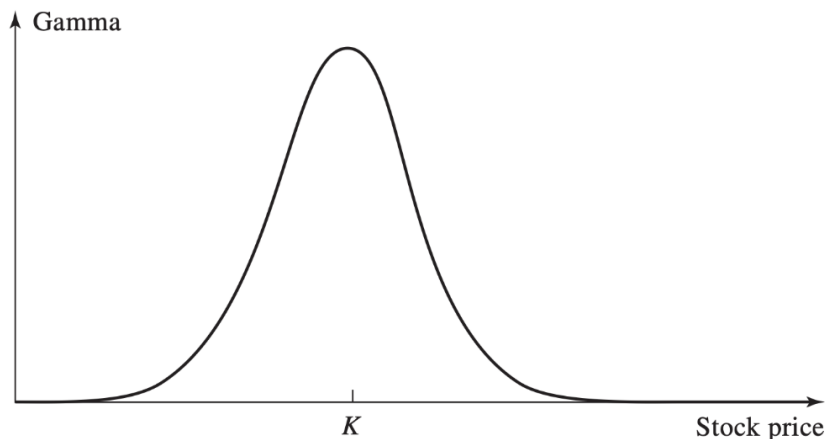


Fig: Relationship between Gamma of an Option and Price of Underlying Stock (K is the Option's Strike Price)

The graph above shows that the gamma of an option is highest when the underlying stock price is near the option's strike price (K), and decreases as the stock price moves away from K . This property is used in practice to manage and neutralize gamma risk in portfolios.

Vega, Theta, and Rho

- **Vega:** Measures the rate of change of the portfolio value with respect to volatility (σ) of the underlying asset.
 - Vega quantifies how sensitive the portfolio is to changes in market volatility.
 - Volatility reflects the uncertainty about the future value of a market variable.
- **Theta:** Measures the rate of change of the portfolio value with respect to time.
 - Indicates how the value of the portfolio erodes as time passes, all else being equal.

- **Rho:** Measures the rate of change of the portfolio value with respect to interest rates.
 - Shows how sensitive the portfolio is to changes in prevailing interest rates.

Taylor Series Expansion for Portfolio Value

The change in portfolio value (ΔP) can be approximated using a Taylor series expansion:

$$\Delta P = \frac{\partial P}{\partial S} \Delta S + \frac{\partial P}{\partial t} \Delta t + \frac{1}{2} \frac{\partial^2 P}{\partial S^2} \Delta S^2 + \frac{1}{2} \frac{\partial^2 P}{\partial t^2} \Delta t^2 + \frac{\partial^2 P}{\partial S \partial t} \Delta S \Delta t + \dots$$

- This expansion expresses how the portfolio value changes with small movements in the underlying asset price (S), time (t), and their higher-order effects.
- Considering very small Δt and the arguments from stochastic calculus showing that ΔS is of order $\sqrt{\Delta t}$, we get, for a delta-neutral portfolio:

$$\Delta P = \Theta \Delta t + \frac{1}{2} \Gamma \Delta S^2$$

where $\Theta = \frac{\partial P}{\partial t}$ and $\Gamma = \frac{\partial^2 P}{\partial S^2}$.

This formula is used to estimate the change in a delta-neutral portfolio over a short time interval, accounting for time decay and the curvature (gamma) effect.

B. Valuation vs Scenario Analysis

Valuation	Scenario Analysis
Estimates expected value of cash flows discounted to present	Explores a full range of possible future situations
Focuses on average or most likely outcomes	Focuses on extreme cases (best/worst) and variability
Provides a single-point estimate	Provides a range of possible outcomes
Useful for standard investment decisions	Useful for risk management and planning under uncertainty

In summary, while valuation gives a baseline estimate under typical conditions, scenario analysis complements it by revealing how sensitive that estimate is to changes in key assumptions, helping to prepare for both expected and unexpected events.

C. Volatility

Volatility and Asset Prices

- It is commonly assumed that asset prices have a constant expected growth rate (μ) and a constant volatility (σ), both expressed per year.
- With continuous compounding, the probability density of the asset price S_T at time T years is:

$$\ln S_T \sim \phi [\ln S_0 + (\mu - \sigma^2/2)T, \sigma^2 T]$$

- This means the logarithm of the asset price at time T is normally distributed with mean $\ln S_0 + (\mu - \sigma^2/2)T$ and variance $\sigma^2 T$. (Comes from the concept of Geometric Brownian Motion (GBM))

Risk-Neutral Valuation

- In a **risk-neutral world**, investors do not require extra compensation for taking risks; the expected return from risky and risk-free investments is the same.
- When valuing derivatives, risk-neutral valuation means we use the price of the underlying asset to calculate the derivative's value, independent of investor risk preferences.
- This approach is a mathematical tool for pricing derivatives; it does not reflect actual investor behavior.
- In practice, risk-neutral valuation provides correct prices for derivatives, even though actual investors do care about risk.
- Moving from risk-neutral to real-world pricing changes both the expected payoff and the discount rate, but these changes offset each other.
- The price of a risky asset is often less than its expected future value, since investors require compensation for risk.

Volatility

- **Volatility** (σ) is the standard deviation of the return provided by a variable per unit of time, typically when the return is expressed using continuous compounding.
- Daily volatility is usually defined as the standard deviation of the proportional change in the variable during a day. This is the standard used in risk management.
- Volatility measures the degree of variation in the price of an asset or market index over time. Higher volatility indicates larger and more unpredictable price swings, while lower volatility reflects more stable prices.
- It is a key input for pricing options and other derivatives, as well as for assessing the risk associated with an asset.

Business Days vs Calendar Days

- Volatility is typically much higher on business days than on non-business days, since trading activity only occurs when markets are open.
- Analysts generally ignore weekends and holidays when calculating and using volatilities, as price changes during these periods are not observable in most markets.
- The standard convention is to assume there are 252 business (trading) days in a year for financial calculations.
- To convert daily volatility to annual volatility, multiply the daily volatility by $\sqrt{252}$. Conversely, to go from annual to daily volatility, divide by $\sqrt{252}$.

What Causes Volatility?

- Researchers analyze daily asset price data to measure volatility.
- Two common measures:
 1. Variance of returns between consecutive trading days.
 2. Variance of returns from Friday close to Monday close.

- Studies show that the variance over a weekend is not simply three times the daily variance, as might be expected.
- This suggests that volatility is not just driven by news or events during non-trading days.
- Research indicates that a significant portion of volatility is caused by trading activity itself.

Are Daily Percentage Changes in Financial Variables Normal?

- In theory, daily changes in financial variables are often assumed to follow a normal distribution.
- For example, under a normal distribution, daily changes exceeding three standard deviations should occur only 0.27% of the time.
- However, empirical data shows that such extreme changes happen more frequently than predicted by the normal distribution.
- This discrepancy arises because, in practice, returns on successive days are not identically distributed—a key requirement for the central limit theorem to hold.
- Market parameters, such as volatility, are not constant over time, leading to distributions with **heavier tails** than the normal distribution.
- As a result, financial returns exhibit more extreme events (outliers) than would be expected if they were truly normal.

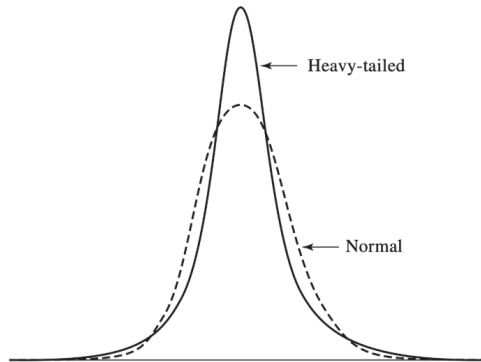


Fig: Comparison of Normal Distribution with a Heavy-Tailed Distribution. Both distributions have the same mean and standard deviation, but the heavy-tailed distribution shows a higher probability of extreme outcomes.

In summary, while the normal distribution is a convenient assumption for modeling daily percentage changes in financial variables, real-world data typically display heavier tails, indicating a higher likelihood of large deviations than the normal model predicts.

Power Law

- The **power law** describes situations where the probability of observing a value greater than x falls off as a power of x :

$$\text{Prob}(v > x) = Kx^{-\alpha}$$

where K and α are constants.

- Power laws are observed in many financial and economic variables, including income, city size, trading volume, and price fluctuations.

- In finance, this means extreme events (like large price changes or crashes) are much more likely than predicted by normal distributions.
- Recognizing power law behavior is crucial for risk management, as it highlights the higher likelihood of rare but significant events.

Estimating Daily Volatility from Historical Data

Variance Rate Formula

The variance rate of a financial variable can be estimated using historical data. The formula for the variance rate is:

$$\sigma_n^2 = \frac{1}{m} \sum_{i=1}^m u_{n-i}^2$$

where u_i is the proportional change in the variable, given by:

$$u_i = \frac{S_i - S_{i-1}}{S_{i-1}}$$

and m is the number of observations.

Derivation and Simplification

Originally, the variance rate was calculated as:

$$\sigma_n^2 = \frac{1}{m-1} \sum_{i=1}^m (u_{n-i} - \bar{u})^2$$

where $\bar{u}$ is the mean of the u_i 's:

$$\bar{u} = \frac{1}{m} \sum_{i=1}^m u_{n-i}$$

However, for financial data (such as daily returns), the following simplifications are often made:

- The mean return $\bar{u}$ is assumed to be zero, since the expected daily change is typically very small compared to the standard deviation.
- The denominator is changed from $m-1$ (unbiased estimator) to m to provide a maximum likelihood estimate.

Key Points

- The simplified variance rate formula assumes the mean return is negligible.
- Using m in the denominator gives a maximum likelihood estimate rather than an unbiased one.
- This approach is commonly used in financial risk management for estimating daily volatility.

Weighting Schemes for Volatility Estimation

Weighted Moving Average

- To estimate volatility (σ_n) on day n , it is common to give more weight to recent data.

- The weighted moving average model is:

$$\sigma_n^2 = \sum_{i=1}^m \alpha_i u_{n-i}^2$$

where α_i is the weight for the observation i days ago, and u_{n-i} is the return.

- The weights satisfy:

$$\sum_{i=1}^m \alpha_i = 1$$

- More recent observations are given higher weights ($\alpha_i < \alpha_j$ for $i > j$).

Incorporating Long-Run Variance

- The model can be extended to include a long-run average variance rate (V_L), giving it weight γ :

$$\sigma_n^2 = \gamma V_L + \sum_{i=1}^m \alpha_i u_{n-i}^2$$

- The weights must sum to unity:

$$\gamma + \sum_{i=1}^m \alpha_i = 1$$

- This is known as an ARCH(m) model.
- Defining $\omega = \gamma V_L$, the model simplifies to:

$$\sigma_n^2 = \omega + \sum_{i=1}^m \alpha_i u_{n-i}^2$$

Key Points

- Weighted schemes allow for more responsive volatility estimates by emphasizing recent data.
- Including a long-run variance component helps stabilize estimates over time.
- These approaches are foundational in financial risk management and volatility modeling.

Exponentially Weighted Moving Average (EWMA)

- The EWMA model updates volatility estimates by giving exponentially decreasing weights to older returns.
- The formula is:

$$\sigma_n^2 = \lambda \sigma_{n-1}^2 + (1 - \lambda) u_{n-1}^2$$

where λ is a constant between 0 and 1, σ_n^2 is the variance estimate, and u_{n-1} is the most recent return.

- A lower λ makes the volatility estimate more responsive to recent changes; a higher λ makes it less responsive.

GARCH(1,1) Model

- The **GARCH(1,1)** model estimates volatility by combining a long-run average variance, the previous day's squared return, and the previous day's variance.
- The formula is:

$$\sigma_n^2 = \gamma V_L + \alpha u_{n-1}^2 + \beta \sigma_{n-1}^2$$

where:

- σ_n^2 : estimated variance for day n
 - V_L : long-run average variance
 - u_{n-1}^2 : squared return from previous day
 - σ_{n-1}^2 : previous day's variance estimate
 - γ, α, β : weights, with $\gamma + \alpha + \beta = 1$
- GARCH(1,1) adapts quickly to recent volatility while anchoring to a long-term average.

GARCH(p,q) Model

- The general **GARCH(p,q)** model uses the most recent p lagged variances and q lagged squared returns:

$$\sigma_n^2 = \omega + \sum_{i=1}^q \alpha_i u_{n-i}^2 + \sum_{j=1}^p \beta_j \sigma_{n-j}^2$$

Mean Reversion in Variance Rates

- In practice, variance rates tend to revert to a long-run average level—this is known as **mean reversion**.
- The **GARCH(1,1)** model captures mean reversion in volatility: variance is pulled back toward a long-term average over time.
- In GARCH(1,1), mean reversion occurs when the sum of the coefficients $\alpha + \beta < 1$.
- The **EWMA** model does not account for mean reversion, as it lacks a long-run average variance term.
- GARCH(1,1) is generally preferred over EWMA when modeling volatility that exhibits mean reversion.

Estimating the Parameters for the Models: Maximum Likelihood Method

- The **maximum likelihood method** estimates model parameters by choosing values that maximize the likelihood of the observed data.
- For a normal random variable with a sequence of observations u_i , the variance ν that maximizes the likelihood is:

$$\nu = \frac{1}{m} \sum_{i=1}^m u_i^2$$

where m is the number of observations.

- This is the maximum likelihood estimator for variance, assuming the mean is zero.
- The unbiased estimator uses $m - 1$ in the denominator instead of m .

Estimating EWMA and GARCH(1,1) Parameters

- Parameters for EWMA and GARCH(1,1) models are typically estimated using the **maximum likelihood method**.

- Assume returns u_i are normally distributed with variance estimate v_i for day i .

- The likelihood to maximize is:

$$\prod_{i=1}^m \left[\frac{1}{\sqrt{2\pi v_i}} \exp \left(-\frac{u_i^2}{2v_i} \right) \right]$$

- Taking logarithms, the log-likelihood simplifies to:

$$\sum_{i=1}^m \left[-\ln(v_i) - \frac{u_i^2}{v_i} \right]$$

- Parameters such as ω, α, β (in GARCH(1,1)) are chosen to maximize this sum, usually via iterative search.

Variance Targeting

- **Variance targeting** is a robust approach for GARCH(1,1) parameter estimation.
- The long-run average variance V_L is set equal to the sample variance from the data.
- The parameter ω is then set as $\omega = V_L(1 - \alpha - \beta)$.
- This reduces the number of parameters to estimate and stabilizes the model.